

21. Merge Two Sorted Lists

PLATFORM

Platform: LeetCode

DIFFICULTY

EASY

DATE

Solved: June 18, 2026

Problem Summary

You are given two sorted linked list :

list1 = 1 -> 2 -> 4

list2 = 1 -> 3 -> 4

Merge them into a single sorted linked list.

Key Insight

Use three pointers : Dummy, curr, list1/list2

Dummy keeps track of the head.

curr builds the merged list.

Compare current nodes of list1 & list2.

Approach

Step1 : Create Dummy Node

Step2 : Compare Nodes

Step3 : Move Current Pointer

Step4 : Attach Remaining Nodes

Step5 : Return Answer

Algorithm

- Create a dummy node.
- Maintain a current pointer.
- While both lists are non-empty:

Compare values
attach the smaller node
move corresponding list pointer
move current pointer

- attach remaining nodes.
- Return dummy.next.

Time Complexity & Space Complexity

TC : $O(n+m)$, n =length of list1 & m =length of list2

SC : $O(1)$

The screenshot displays the LeetCode problem page for "21. Merge Two Sorted Lists". The problem description states: "You are given the heads of two sorted linked lists list1 and list2. Merge the two lists into one sorted list. The list should be made by splicing together the nodes of the first two lists. Return the head of the merged linked list." Example 1 shows list1 = [1,2,4] and list2 = [1,3,4] merging to [1,1,2,3,4,4]. Example 2 shows list1 = [] and list2 = [] merging to []. Example 3 shows list1 = [] and list2 = [0] merging to [0]. The solution code is in Java, using a dummy node and a current pointer to merge the two lists. The test results show the solution is "Accepted" with a runtime of 0 ms and memory usage of 44.06 MB.

21. Merge Two Sorted Lists Solved

Easy Topics Companies

You are given the heads of two sorted linked lists `list1` and `list2`.

Merge the two lists into one sorted list. The list should be made by splicing together the nodes of the first two lists.

Return the head of the merged linked list.

Example 1:

Input: list1 = [1,2,4], list2 = [1,3,4]
Output: [1,1,2,3,4,4]

Example 2:

Input: list1 = [], list2 = []
Output: []

Example 3:

Input: list1 = [], list2 = [0]
Output: [0]

Constraints:

- The number of nodes in both lists is in the range $[0, 50]$.

Accepted 208 / 208 testcases passed
Sahil Khan submitted at Jun 18, 2026 23:25

Runtime
0 ms | Beats 100.00%

Memory
44.06 MB | Beats 87.11%

Code | Java

```
1 /**
2  * Definition for singly-linked list.
3  * public class ListNode {
4  *     int val;
5  *     ListNode next;
6  *     ListNode() {}
7  *     ListNode(int val) { this.val = val; }
8  *     ListNode(int val, ListNode next) { this.val =
9  * 
```

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

list1 = [1,2,4]

list2 = [1,3,4]